

Background and Conflication

The understanding behind the mRNA COVID-19 vaccine starts with the delivery of mRNA via lipid nanoparticles to the cells. The next stage in this process is the translation of the delivered mRNA in the cells' cytoplasm. The translated material will serve as a template for the creation of the SARS-CoV-2 spike protein. Afterward, it triggers the immune system, including B cells and T cells, and antibody response. The reason for choosing the spike protein is the fact that it is the main surface protein that binds to ACE2 receptors during the infection of SARS-CoV-2, and this binding helps the virus enter and infect cells.

As the result of using such vaccines on large populations, some rare cardiovascular complications have been revealed. Myocarditis, pericarditis, and thrombosis with thrombocytopenia syndrome or VITT/TTS are among the most rare cardiovascular diseases prominently reported post certain COVID - 19 vaccinations . Myocarditis and pericarditis have rarely been associated with mRNA vaccines but affect predominantly young men. In turn, VITT is mostly associated with adenoviral vector vaccines. One more thing to keep in mind is the fact that the presence of any condition after vaccination does not mean that the vaccine caused it.

Biological Understanding and posed mechanisms

Lipid nanoparticles provide a barrier which protects the mRNA by encapsulating it to combat enzymatic degradation and successfully enter within the cells through the process called endocytosis. Once inside the cells, the mRNA goes past the endosome and reaches the cytoplasm, where it is translated to produce spike protein. The spike protein is detected by the immune system resulting in activation of dendritic cells, helper T cells, cytotoxic T cells, and B cells. As a result, production of antibodies and immunity memory occurs. LNPs might have an adjuvant effect, meaning that they stimulate the innate immune system and affect the intensity of the immune response to the vaccine.

The exact biological understanding of myocarditis and pericarditis being caused by vaccination is still something not fully uncovered. Possible mechanisms include hyperactivation of the immune system, release of inflammatory cytokines, T-cell-dependent inflammation, molecular mimicry, and possibly individual genetic and/or hormonal predisposition. VITT develops via a different mechanism. In this case, the antibodies are produced against platelet factor 4 (PF4) and platelet activation takes place. It results in thrombus formation and decrease in platelet counts.

Significant Cardiovascular Complications and Risk Factors

The main rare diseases that become relevant post vaccination can be Myocarditis, Pericarditis, and VITT. The disease, myocarditis, focuses on the inflammation of the heart muscle. The disease pericarditis is inflammation of the sac surrounding the heart. Both diseases share common symptoms of chest pain, dyspnea, and palpitations. Post-vaccination myocarditis and

pericarditis have been rarely observed but are significantly seen among teenagers, young adults and adolescent males, usually around seven days following the second dose of an mRNA-based vaccine. The actual mechanism causing these problems remains unclear; some factors lead to it possibly being due to the response of the immune system and other factors such as sex and age.

The main difference between VITT and other post-vaccination side effects is that it involves unusual thrombotic occurrences with low platelet levels without inflammation in the heart. The major causes of VITT include adenoviral-vector vaccines against the coronavirus such as the Janssen and AstraZeneca vaccines. In essence, the risks associated with each disease are different. For example, myocarditis and pericarditis are mainly found in younger males who have received an mRNA-based vaccine while VITT/TTS is mostly associated with exposure to certain adenoviral-vector vaccines.

Citations

Block, J. P. (2022). Cardiac Complications After SARS-CoV-2 Infection and mRNA COVID-19 Vaccination — PCORnet, United States, January 2021–January 2022. *MMWR. Morbidity and Mortality Weekly Report*, 71. <https://doi.org/10.15585/mmwr.mm7114e1>

CDC. (2025, February 6). *Grading of Recommendations, Assessment, Development, and Evaluation (GRADE): Updated COVID-19 vaccine (2023-2024 Formulation)*. Advisory Committee on Immunization Practices (ACIP). <https://www.cdc.gov/acip/grade/covid-19-2023-2024-Monovalent.html>

Wong, H.-L., Hu, M., Zhou, C. K., Lloyd, P. C., Amend, K. L., Beachler, D. C., Secora, A., McMahonill-Walraven, C. N., Lu, Y., Wu, Y., Ogilvie, R. P., Reich, C., Djibo, D. A., Wan, Z., Seeger, J. D., Akhtar, S., Jiao, Y., Chillarige, Y., Do, R., ... Anderson, S. A. (2022). Risk of myocarditis and pericarditis after the COVID-19 mRNA vaccination in the USA: a cohort study in claims databases. *The Lancet*, 399(10342), 2191–2199. [https://doi.org/10.1016/S0140-6736\(22\)00791-7](https://doi.org/10.1016/S0140-6736(22)00791-7)

Han, X., Alameh, M.-G., Butowska, K., Knox, J. J., Lundgreen, K., Ghattas, M., Gong, N., Xue, L., Xu, Y., Lavertu, M., Bates, P., Xu, J., Nie, G., Zhong, Y., Weissman, D., & Mitchell, M. J. (2023). Adjuvant lipidoid-substituted lipid nanoparticles augment the immunogenicity of SARS-CoV-2 mRNA vaccines. *Nature Nanotechnology*, 18(9), 1105–1114. <https://doi.org/10.1038/s41565-023-01404-4>

Zhang, Y., Wang, J., Xing, H., Chao, L., Zha, W., Dong, S., Jiang, Y., & Li, X. (2023). Enhanced immunogenicity induced by mRNA vaccines with various lipid nanoparticles as carriers for SARS-CoV-2 infection. *J. Mater. Chem. B*, 11(31), 7454–7465. <https://doi.org/10.1039/d3tb00303e>

Locci, M., & Pardi, N. (2026). Immunological mechanisms of mRNA vaccines for infectious diseases. *Nature*, 654(8120), 892–901. <https://doi.org/10.1038/s41586-026-10599-0>

